MILITARY SPECIFICATION

CAPACITORS, FIXED, MICA DIELECTRIC,

HIGH RELIABILITY, GENERAL SPECIFICATION FOR

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for high reliability, mica dielectric, fixed capacitors for space, missile, and other high reliability applications. Capacitors covered by this specification may be used in critical applications where the utmost in stability and dependability is required and where variations in capacitance with respect to temperature, voltage, frequency, and life cannot be tolerated (see 6.1). An acceleration factor of 25:1 has been used to relate life test data obtained at 150 percent of rated voltage at rated temperature to rated voltage at rated temperature.

1.2 Classification.

1.2.1 Military part number. The capacitors specified herein (see 3.1) shall be identified by a military part number which shall consist of the basic number of the military specification sheet followed by a series of coded characters. Each military specification sheet covers a different capacitor style. The coded number shall provide information concerning the capacitors characteristic, working voltage, capacitance value, and capacitance tolerance. The military part number shall be in the following form as indicated:

M87164	A	01	C	D	1R0	D
Military specification number	Modification 1.2.1.1	Slash sheet number	Characteristic 1.2.1.2	Voltage 1.2.1.3	Capacitance value 1.2.1.4	Capacitance tolerance 1.2.1.5

1.2.1.1 Modification. The modification letter is used to indicate the latest characteristic of the part in the specification sheet. This letter shall be changed under direction of the preparing activity.

1.2.1.2 Characteristic. The characteristic is identified by a single letter which indicates the relative stability of the capacitor with temperature change, in accordance with table I.

TABLE I. Characteristic.

Symbol	Temperature Coefficient	Capacitance drift
	Parts/million/°C	
C	-200 to +200	±(0.5 percent +0.1 pF)
E	-20 to +100	±(0.1 percent +0.1 pF)
F	0 to +70	±(0.05 percent +0.1 pF)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: SD/ALM, P.O. Box 92960, Worldway Postal Center, Los Angeles, CA 90009, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2.1.3 Voltage. The rated voltage for continuous operation at +125°C is identified by a single letter as shown in table II.

TABLE II. Rated voltage.

Symbol	Rated voltage
	Volts, dc
A	100
C	300
D	500

1.2.1.4 Capacitance. The nominal capacitance value, expressed in picofarads (pF) is identified by a three-digit number; the first two digits represent significant figures and the last digit specifies the number of zeros to follow. When the nominal value is less than 10 pF, the letter "R" shall be used to indicate the decimal point and the succeeding digit shall represent a significant figure. For example, 1R0 indicates 1.0 pF.

1.2.1.5 Capacitance tolerance. The capacitance tolerance is identified by a single letter in accordance with table III.

TABLE III. Capacitance tolerance.

Symbol	Capacitance tolerance
	±
D	0.5 pF
F	1 percent
G	2 percent
J	5 percent

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-39001/5 - Capacitors, Fixed, Mica Dielectric, Established Reliability; Styles CMR04, CMR05, CMR06, CMR07 and CMR08.
MIL-C-39028 - Capacitors, Packaging of.
MIL-C-87164/1 - Capacitors, Fixed, Mica Dielectric High Reliability Style CMS04.
MIL-C-87164/2 - Capacitors, Fixed, Mica Dielectric, High Reliability Style CMS05.
MIL-C-87164/3 - Capacitors, Fixed, Mica Dielectric, High Reliability Style CMS06.

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-790 - Reliability Assurance Program for Electronic Parts Specifications.
MIL-STD-810 - Environmental Test Methods.
MIL-STD-1276 - Leads For Electronic Component Parts.
MIL-STD-1285 - Marking of Electrical and Electronic Parts.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern (see 6.1).

3.2 Qualification. The capacitors furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids. Distributors shall not be used (see 4.4 and 6.3).

3.3 Reliability and baseline documentation. The capacitors supplied to this specification shall conform to the requirements of MIL-STD-790 and the requirements specified herein. Prior to granting of qualification approval, baseline documentation (see appendix A) shall be approved by the qualifying activity. No changes in material, design, construction, or testing of the products qualified, except as specifically allowed by the baseline, shall be made without changing documentation and receiving approval of the qualifying activity.

3.4 Material. The material shall be as specified herein. However, when a definite material is not specified, a material shall be used that will enable the capacitors to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.4.1 Insulating, impregnating, and sealing compounds. Insulating, impregnating, and sealing compounds, including varnishes, waxes, and the like, shall be suitable for each particular application. Compounds shall preserve the electrical characteristics of the insulation to which applied. The compound, either in the state of its original application or as a result of cracking or aging, shall have no adverse effect on the performance of the capacitor.

3.4.2 Terminal leads. Lead wires shall be tin-lead coated copper (Type C of MIL-STD-1276) or tin-lead coated copper clad steel (Type W of MIL-STD-1276) of the length and diameter specified (see 3.1).

3.4.3 Dielectric. The mica dielectric thickness and safety margins shall be approved by the baseline.

3.5 Design and construction. The capacitors shall be of the materials, design, construction and physical dimensions specified (see 3.1) and in accordance with baseline documentation (see 3.3).

3.5.1 Body enclosure. The capacitors shall be effectively sealed against the entry of moisture, and the elements shall be mounted so as to prevent injurious movement in the capacitor body enclosures. The capacitor elements shall be completely enclosed by the body enclosure.

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3.5.2 Connections. All electrodes (foils, clips and leads) used as terminations for the silvered mica conductor must be mechanically and electrically bonded together. Foils (and solder if used) must have a minimum melting point of 260°C. Solder, if used, for bonding the lead wires to the clips will have a minimum melting point of 182°C and will be of the same alloy as the tin-lead coating on the lead wire.

3.6 Dielectric withstanding voltage. The capacitors shall withstand the direct current (dc) potential specified in 4.7.2 without damage, arcing, or breakdown.

3.7 Barometric pressure. The capacitors shall withstand the dc potential specified in 4.7.3 without damage, arcing, or breakdown.

3.8 Insulation resistance.

3.8.1 At room ambient temperature. When measured as specified in 4.7.4.1, the insulation resistance shall be not less than the applicable value specified on figure 1.

3.8.2 At high ambient temperature. When measured as specified in 4.7.4.2, the insulation resistance shall be not less than the applicable value specified on figure 1.

3.9 Capacitance. When measured as specified in 4.7.5, the capacitance shall be within the tolerance designated by the part number (see 3.1).

3.10 Dissipation factor. When measured as specified in 4.7.6, the dissipation factor shall not exceed the applicable value shown on figure 2.

3.11 High voltage stabilization. When the capacitors are tested as specified in 4.7.7, there shall be no evidence of damage, arcing, or breakdown; and the insulation resistance shall be not less then the applicable value specified on figure 1.

3.12 Solderability. When the capacitors are tested as specified in 4.7.8, the dipped surface of the lead shall be at least 95 percent covered with continuous new solder coating. The remaining 5 percent of the lead surface shall show only small pinholes or voids. These shall not be concentrated in one area. No individual view of the dipped surface shall show less than 95 percent coverage. Bare base metal and areas where the solder dip failed to cover the original coating are indications of poor solderability, and shall be cause for failure. In case of dispute, the percent of coverage with pinholes or voids shall be determined by actual measurement of these areas, as compared to the total area.

3.13 Vibration, high frequency. When the capacitors are tested as specified in 4.7.9, there shall be no intermittent contacts of 0.5 milliseconds (ms) or greater duration, or momentary arcing, or other indication of breakdown, nor shall there be any open- or short-circuiting or evidence of mechanical damage.

3.14 Temperature coefficient and capacitance drift. When measured as specified in 4.7.10, the temperature coefficient and capacitance drift shall be within the limits specified in table I for the characteristic listed (see 3.1).

3.15 Thermal shock and immersion. When tested as specified in 4.7.11, the capacitors shall meet the following requirements:

Dielectric withstanding voltage - - - -	As specified in 3.6.
Insulation resistance - - - - - - - - -	Shall be not less than 30 percent of initial requirement (see 3.8).
Capacitance - - - - - - - - - - - - - -	The capacitance change shall not exceed 1 percent or 1 pF, whichever is greater, from the reference measurement (see 4.3.2.1).
Dissipation factor- - - - - - - - - - -	Shall not exceed 150 percent of the initial requirement specified in 3.10.
Visual examination- - - - - - - - - - -	No visible evidence of deterioration, permanent damage to leads or case, or corrosion on leads.

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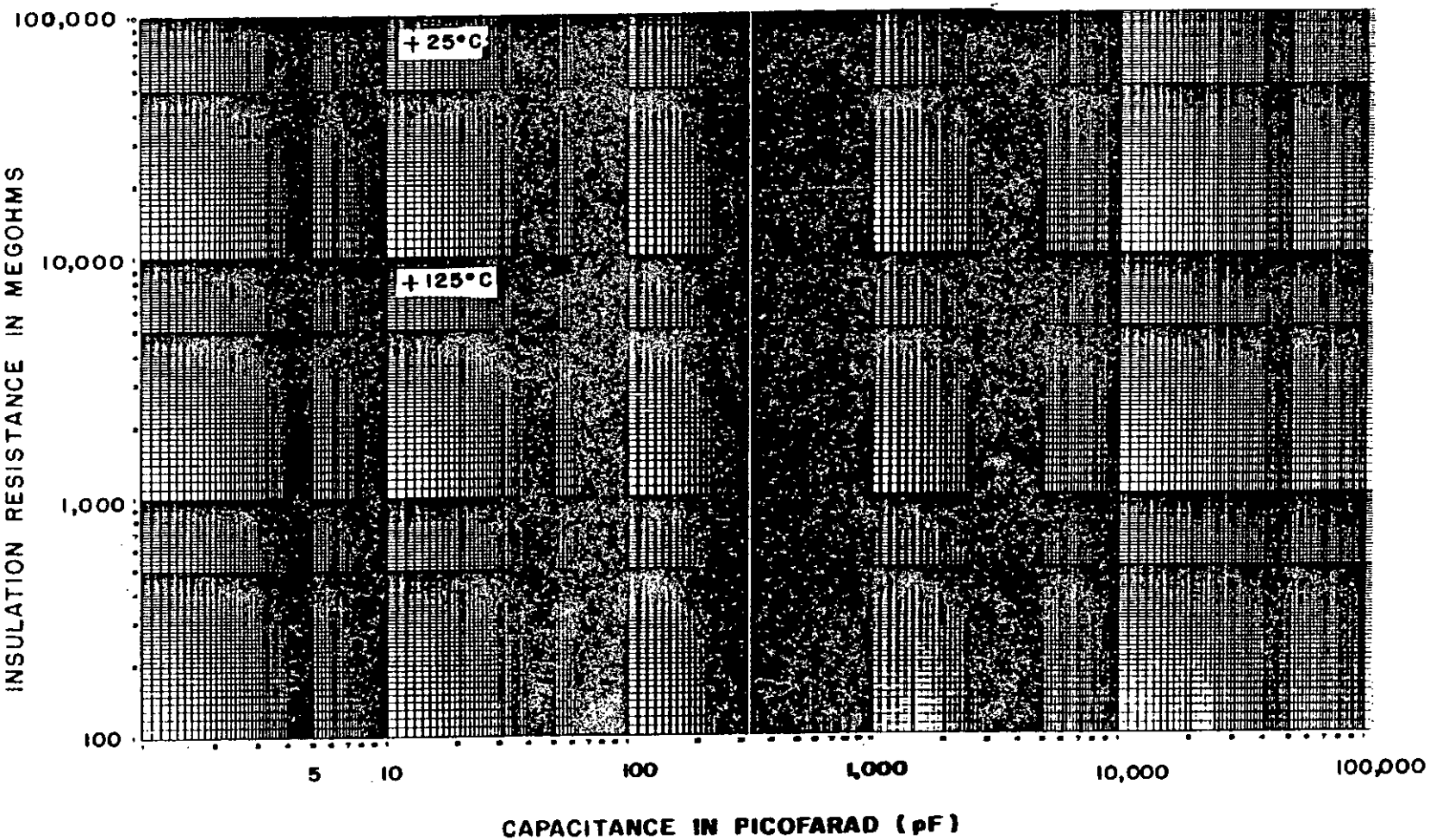


FIGURE 1. Insulation resistance vs capacitance at test temperatures.

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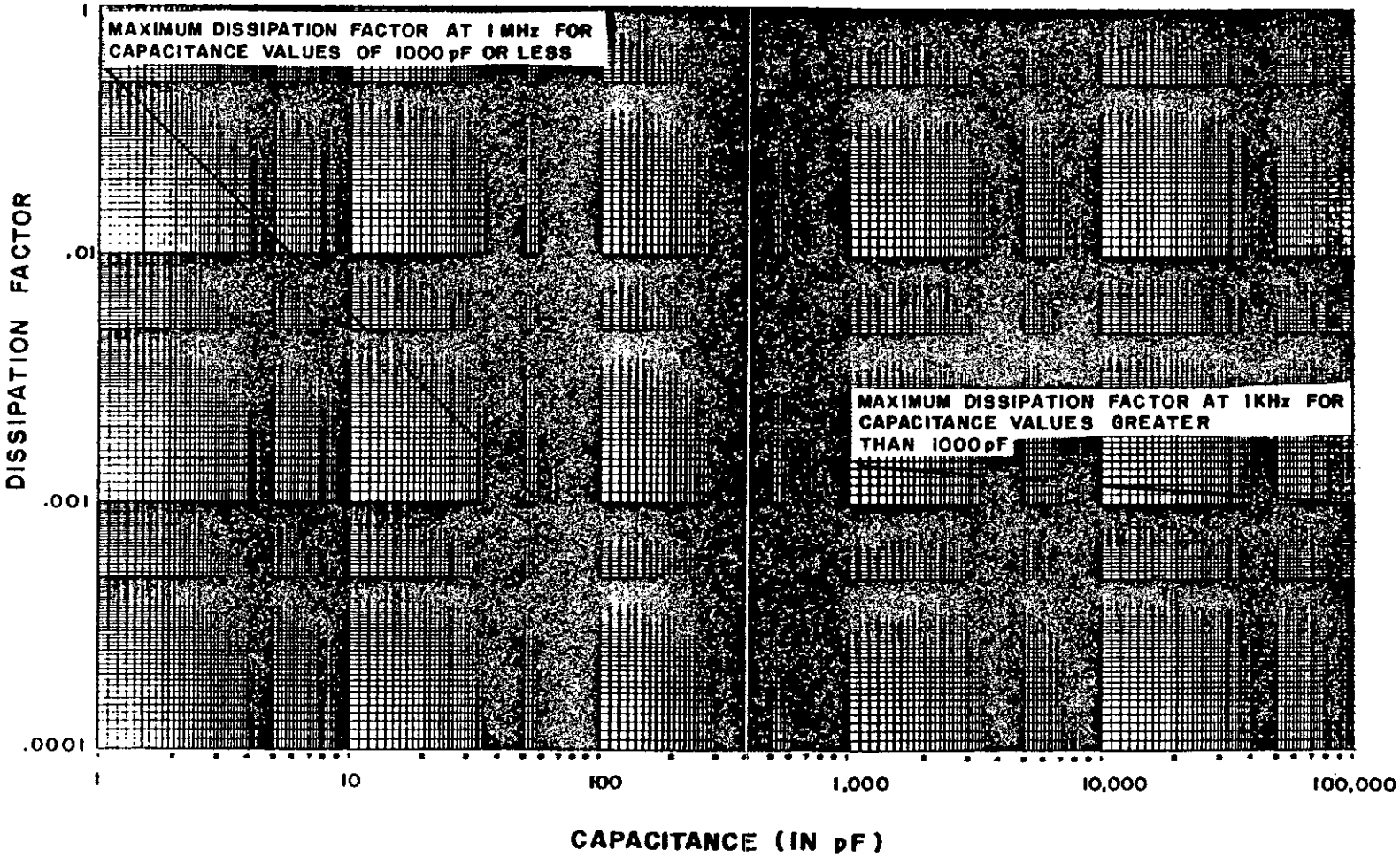


FIGURE 2. Capacitance vs dissipation factor.

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3.16 Shock, specified pulse. When the capacitors are tested as specified in 4.7.12, there shall be no intermittent contacts of 0.5 milliseconds or greater duration, or momentary arcing, or other indication of breakdown, nor shall there be any open- or short-circuiting or evidence of mechanical damage.

3.17 Terminal strength. When the capacitors are tested as specified in 4.7.13, the terminals shall not loosen or rupture and no other damage shall result. Chipping of the resinous coating on the leads, in the "R" dimension area only (see 3.1), shall not be considered as a failure.

3.18 Moisture resistance. When tested as specified in 4.7.14, capacitors shall meet the following requirements:

Dielectric withstanding voltage - - - -	As specified in 3.6.
Insulation resistance - - - - - - - - - -	Shall be not less than 25 percent of initial requirement (see 3.8).
Capacitance - - - - - - - - - - - - - - -	The capacitance change shall not exceed 1 percent or 1 pF, whichever, is greater, from the reference measurement (see 4.3.2.1).
Dissipation factor- - - - - - - - - - - -	Shall not exceed 150 percent of the initial requirement specified in 3.10.
Visual examination- - - - - - - - - - - -	No visible evidence of deterioration, permanent damage to the leads or case, nor corrosion on the leads.

3.19 Life. When tested as specified in 4.7.15, capacitors shall meet the following requirements:

Dielectric withstanding voltage - - - -	As specified in 3.6.
Insulation resistance - - - - - - - - - -	Shall not be less than the applicable value specified on figure 1.
Capacitance - - - - - - - - - - - - - - -	The capacitance change shall not exceed ±1 percent or ±1 pF, whichever is greater, of the initial measurement.
Dissipation factor- - - - - - - - - - - -	Shall not be greater than 150 percent of initial requirement (see 3.10).
Visual examination- - - - - - - - - - - -	No visible evidence of deterioration, permanent damage to leads or case, arcing, or breakdown.

3.20 Fungus. The manufacturer shall certify that all external materials are fungus resistant or shall perform the test specified in 4.7.16. When the capacitors are tested as specified in 4.7.16, examination shall disclose no evidence of fungus growth on the external surface of the capacitor.

3.21 Resistance to solvents. When the capacitors are tested as specified in 4.7.17, marking shall remain legible and shall not smear or rub off. In addition, there shall be no visible indication of damage or deterioration to the capacitor body.

3.22 Resistance to soldering heat. When tested as specified in 4.7.18, capacitors shall meet the following requirements:

Insulation resistance - - - - - - - - - -	Shall not be less than the initial requirement (see 3.8).
Capacitance - - - - - - - - - - - - - - -	The capacitance change shall not exceed ±5 percent or ±5 pF, whichever is greater, from the initial measurement.
Dissipation factor- - - - - - - - - - - -	Shall not exceed initial requirement.

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3.23 Marking. Marking of capacitors shall conform to method I of MIL-STD-1285, and unless otherwise specified (see 3.1), shall include the type designation, "JAN" brand, trademark, source code, date code (inspection lot of 4.6.1), rated voltage, capacitance tolerance, and highest rated temperature. Capacitors shall be marked as specified (see 3.1).

3.23.1 JAN and J marking. The United States Government has adopted, and is exercising legitimate control over the certification marks "JAN" and "J", respectively, to indicate that items so marked or identified are manufactured to, and meet all the requirements of military specifications. Accordingly, items acquired to, and meeting all of the criteria specified herein and in applicable detail specifications, shall bear the certification mark "JAN" except that items too small to bear the certification mark "JAN" shall bear the letter "J". The "JAN" or "J" shall be placed immediately before the part number except that if such location would place a hardship on the manufacturer in connection with such marking, the "JAN" or "J" may be located on the first line above or below the part number. Items furnished under contracts or purchase orders which either permit or require deviation from the conditions or requirements specified herein or in applicable detail specifications shall not bear "JAN" or "J". In the event an item fails to meet the requirements of this specification and the applicable specification sheets or detail specifications, the manufacturer shall remove the "JAN" or the "J" from the sample tested and also from all items represented by the sample. The "JAN" or "J" certification mark shall not be used on products acquired to contractor drawings or specifications. The United States Government has obtained Certificate of Registration No. 504,860 for the certification mark "JAN".

3.24 Workmanship. The capacitors shall be processed in such a manner as to be uniform in quality and shall be free from pits, corrosion, cracks, rough edges, and other defects that will affect life, serviceability, or appearance.

3.24.1 Soldering. Where soldering is employed in the construction of capacitors, only fluxes with low corrosive properties shall be used unless it can be shown that corrosive elements have been satisfactorily removed after soldering. All excess flux and solder shall be removed. Solder shall not be used primarily for obtaining mechanical strength. Electrical connections shall be electrically continuous after soldering and, where possible, mechanically secure before soldering.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Reliability assurance program. A reliability assurance program shall be established and maintained in accordance with MIL-STD-790. Evidence of such compliance shall be verified by the qualifying activity of this specification as a prerequisite for qualification and continued qualification.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

(a) Qualification inspection (see 4.4).

(b) Retention of qualification (see 4.5).

(c) In-process inspection and quality conformance inspection (see 4.6).

4.3 Inspection conditions and methods.

4.3.1 Conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of MIL-STD-202.

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4.3.2 Methods.

4.3.2.1 Reference measurements. When requirements are based on comparative measurements made before and after conditioning, the reference measurement shall be considered the last measurement made at 25°C ±5°C prior to conditioning. Unless reference measurements have been made within 30 days prior to the beginning of conditioning, they shall be repeated.

4.3.3 Power supply. The power supply used for life testing shall have a regulation of ±2 percent or less of the applicable applied test voltage. The power supply used for dc leakage current measurements shall be stabilized to at least ±100 parts per million. No voltage fluctuations shall occur during measurements that would produce a variation in the current measurement.

4.4 Qualification inspection. Qualification inspection shall be performed at a laboratory acceptable to the Government (see 6.3), on sample units produced with equipment and procedures normally used in production. The decision as to whether or not the product will be included in the QPL will be made after successful completion of the 2,000-hour life test.

4.4.1 Sample size. The number of sample units comprising a sample of the capacitors to be submitted for qualification inspection shall be as specified in table IV, or in appendix A to this specification. The sample shall be taken from a production run and shall be produced with equipment and procedures normally used in production. Each capacitor style shall be qualified separately.

4.4.2 Test routine. Sample units shall be subjected to the qualification inspection specified in table IV, in the order shown. All sample units shall be subjected to the group I and group II tests. These sample units shall then be divided as shown in table IV for groups III to VI, inclusive, and subjected to the tests for that particular group.

4.4.3 Failures. Failures in excess of those allowed in table IV shall be cause for refusal to grant qualification approval.

4.4.4 Qualification. A manufacturer shall be considered qualified for the CMS04, CMS05, and CMS06 if he is presently qualified to the R failure level of MIL-C-39001/5 for the CMR04, CMR05, and CMR06 and meets the following conditions:

a. Passes the qualifying activity audit for CMS style capacitors (see appendix B).

b. Successfully completes group C inspection (table VII) for the lowest (lowest value need not be less than 47 pF) and highest capacitance values for each different style manufactured.

4.5 Retention of qualification. To retain qualification for capacitors specified in this specification, the manufacturer shall forward a report at 6-month intervals to the qualifying activity. The qualifying activity shall establish the initial reporting date. The report shall include the following:

a. A list of all device types for which the manufacturer has qualification, showing the total number of lots completing quality conformance inspection during the reporting period.

b. Certification that the manufacturer still has the capabilities and facilities necessary to produce these items to the qualified levels.

c. The number of lots which failed lot by lot testing or periodic testing (group B or group C).

d. For failed lots (group B and group C) the following information is required:

(1) Inspection lot identification code and part number(s) in the failed lot.
(2) The percent defective of voltage conditioning.

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(3) The results of groups A, B, and C testing (i.e., pass or fail).
(4) For failed subgroups, the number and mode of all failures.
(5) All lots withdrawn shall be counted as failed lots in the report.
(6) Identification of the lots (by inspection lot identification code, part number, and date of completion of group B testing).

TABLE IV. Qualification inspection.

Inspection	Requirement paragraph	Method paragraph	Number of sample units to be inspected	Number of defectives allowed 1/
Group I 2/				
Dielectric withstanding voltage	3.6	4.7.2	125	0
High voltage stabilization	3.11	4.7.7		
Group II 2/				
Visual and mechanical examination 3/	3.1,3.4 to 3.5.2 incl 3.23 to 3.24.1 incl	4.7.1	125 4/	0
Barometric pressure	3.7	4.7.3		
Insulation resistance	3.8	4.7.4		
Capacitance	3.9	4.7.5		
Dissipation factor	3.10	4.7.6		
Group III				
Solderability	3.12	4.7.8	18	1
Vibration, high frequency	3.13	4.7.9		
Temperature coefficient and capacitance drift	3.14	4.7.10		
Thermal shock and immersion	3.15	4.7.11		
Group IV				
Shock, specified pulse	3.16	4.7.12	18	1
Terminal strength	3.17	4.7.13		
Moisture resistance	3.18	4.7.14		
Group V				
Life (accelerated conditions)	3.19	4.7.15.1	79	1
Group VI				
Fungus 5/	3.20	4.7.16	5	0
Resistance to solvents	3.21	4.7.17	5	
Resistance to soldering heat	3.22	4.7.18		

1/ A sample unit having one or more defects shall be considered as a single defective.
2/ Nondestructive tests.
3/ Marking defects are based on visual examination only and shall be charged only for illegible, incomplete, or incorrect marking.
4/ Sample units shall be those that have passed group I inspection.
5/ Certification of fungus resistance may be substituted for testing.

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The report shall also include certification that the design and construction of the product was verified and found to be identical to the style for which qualification approval was valid. If the summary of test results indicates nonconformance to the requirements of this specification and corrective action acceptable to the qualifying activity has not been taken, action may be taken to remove the failing product from the qualified products list. Failure to submit the summary report within 30 days after the end of the 12-month period may result in loss of qualification for products within that capacitor's group. In addition to the periodic submission of inspection data, the manufacturer shall within 30 days notify the qualifying activity at any time during the 12-month period that the inspection data indicates inability of the qualified product to meet the requirements of this specification. If during two consecutive reporting periods there has been no quality conformance inspection of product within a qualified technology, the manufacturer may, at the discretion of the qualifying activity, be required to submit a representative product(s) of that qualified technology to complete testing in accordance with the qualification requirements. In this case the product submitted must comply with the baseline.

4.6 In-process inspection and quality conformance inspection.

4.6.1 Inspection lot. An inspection/production lot shall be of one design as specified in the baseline documentation. As a minimum requirement, the lot shall consist of all the capacitors of a single nominal capacitance/voltage rating of one design, and processed as a single lot continuously through all manufacturing steps on the same equipment, to the same baseline/revision and identified with the same date and lot code designation. The lot may contain all available capacitance tolerances for the nominal capacitance value as well as adjacent values. In addition, the lot shall conform to the following:

a. Materials such as silver, foil, terminations, solder, lead material, etc., shall be uniform in quality and from a single supplier.

b. The lot code number shall be assigned at or before assembly.

c. Traceability to the operator and the date each operation was performed shall be maintained.

d. Lot identity shall be maintained from the time the lot is assembled to the time it is accepted.

4.6.2 In-process inspection. Each production lot of parts shall be inspected to an approved baseline procedure for the following:

a. Silvered mica shall be 100% visually inspected under 4X minimum magnification prior to assembly.

b. Pre-encapsulation inspection shall be performed on 100% of the parts under 4X minimum magnification. Lead attachment and general workmanship shall be examined.

4.6.2.1 Rework. Rework is not allowed on any operation except for marking and lead rework or as accepted by the baseline.

4.6.3 Inspection of product for delivery. Inspection of product for delivery shall consist of group A and group B inspections.

4.6.3.1 Group A inspection. Group A inspection shall consist of the inspection specified in table V. Failure to meet a 5 percent defective allowable (PDA), shall be cause for lot rejection and the lot shall not be resubmitted. A summary of group A test data shall be furnished the acquiring activity with each lot.

4.6.3.2 Group B (lot by lot) inspection. Group B inspection shall consist of the tests specified in table VI and shall be performed on sample units from lots that have been subjected to and have passed group A inspection. Copies of group B test data shall be furnished the acquiring activity with each lot. Parts shall not be shipped until the conclusion of 250 hours of life test and must meet the requirements of table VI (waivers to ship prior to completion of group B may be granted in writing by the acquiring activity). Group B samples and data shall be maintained by the manufacturer for 3 years.

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TABLE V. Group A inspection.

Inspection	Requirement paragraph	Method paragraph	Sample size	PDA
Thermal shock	3.15	4.7.11.1	100%	
High voltage stabilization	3.11	4.7.7	100%	
Dielectric withstanding voltage	3.6	4.7.2	100%	5%
Insulation resistance at 25°C	3.8.1	4.7.4.1	100%	
Capacitance 1/	3.9	4.7.5	100%	1/ 3/
Dissipation factor	3.10	4.7.6	100%	2/ 3/
Visual and mechanical	3.1, 3.4 to 3.5 incl 3.23 to 3.24.1 incl	4.7.1	AQL 1.0	

1/ Capacitance values outside the initial limits by 1 percent or 1 pF, whichever is greater, shall be used for PDA lot rejection.
2/ Dissipation factor greater than 130 percent of the initial limit shall be used for PDA lot rejection.
3/ All capacitors delivered to this specification shall meet the nominal requirements.

4.6.4 Periodic inspection. Periodic inspection shall consist of group C inspection. Except where the results of this specification show noncompliance with the applicable requirements, delivery of products which have passed group A and group B inspections shall not be delayed pending the results of these periodic inspections. Group C data shall be submitted to the qualifying activity every 6 months.

4.6.4.1 Group C inspection. Group C inspection shall consist of the tests specified in table VII in the order shown. Samples having group C testing shall be selected from lots that have passed group A and have been submitted for group B inspection. Samples shall be selected every 2 months. Capacitance values manufactured during the period shall be represented as far as practical in the approximate ratio of production. Group C samples and data shall be maintained by the manufacturer for 3 years.

4.6.4.2 Failure analysis. If any of the sample units subjected to the group C life test fail, a detailed failure analysis shall be conducted to establish the cause of failure and the corrective actions. A failure is categorized as "lot oriented" if its occurrence is apparently related to an identified lot or lots. A failure is categorized as "not lot oriented" if its occurrence is random and it cannot be related to a specific lot or lots. Each failure is further identified as "screenable" or "not screenable" from the completed production items. If the failure analysis shows that the failure mechanism is "screenable", the entire failed lot may be screened and the group C life test shall be repeated. If a failure occurs during the second group C life test, the entire production lot shall be rejected. If the failure mechanism is "screenable", all prior and subsequent production lots that may contain the identified failure mechanism shall also be screened. If the failure mechanism is "lot oriented" and "not screenable", all production lots that may contain the identified failure mechanism shall be rejected unless other disposition is directed by the contracting officer.

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TABLE VI. Group B lot by lot inspection.

Inspection	Requirement paragraph	Test method paragraph	Sample size	Accept
Subgroup 1				
Insulation resistance at +25°C and +125°C	3.8	4.7.4	20	0
Moisture resistance	3.18	4.7.14		1
Subgroup 2				
Life test (250 hours accelerated)	3.19	4.7.15.2.1	20	0

TABLE VII. Group C periodic inspection.

Inspection	Requirement paragraph	Test method paragraph	Sample size	Accept
Subgroup 1				
Solderability	3.12	4.7.8	20	1
Vibration, high frequency	3.13	4.7.9		
Subgroup 2				
Temperature coefficient and capacitance drift	3.14	4.7.10	20	1 1/
Thermal shock (25 cycles) and immersion	3.15	4.7.11		
Subgroup 3				
Shock, specified pulse	3.16	4.7.12	20	1
Terminal strength	3.17	4.7.13		
Subgroup 4				
Life test (2000 hours accelerated)	3.19	4.7.15.2.2	20	0 2/

1/ One failure allowed into the adjoining characteristic. If capacitance drifts more than the adjoining characteristic, the lot shall be recalled.

2/ See failure analysis.

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4.6.4.3 Disposition of sample units. Sample units which have been subjected to group C inspection shall not be delivered on the contract or purchase order.

4.6.4.4 Noncompliance. If a sample fails to pass group C inspection, the manufacturer shall take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes, etc., and which are considered subject to the same failure. Acceptance of the product shall be discontinued until corrective action, acceptable to the Government, has been taken. After the corrective action has been taken, group C inspection shall be repeated on additional sample units (all inspection, or the inspection which the original sample failed, at the option of the Government). In the event of failure after reinspection, information concerning the failure and corrective action taken shall be furnished to the cognizant inspection activity and the qualifying activity.

4.6.5 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the requirements of MIL-C-39028.

4.7 Methods of inspection.

4.7.1 Visual and mechanical examination. Capacitors shall be examined to verify that the materials, design, construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements (see 3.1, 3.4 to 3.5.2 inclusive, and 3.23 to 3.24.1 inclusive).

4.7.2 Dielectric withstanding voltage (see 3.6). Capacitors shall be tested in accordance with method 301 of MIL-STD-202. The following details shall apply:

a. Magnitude and nature of test voltage: 200 percent of rated dc voltage.

b. Duration of application of test voltage: Not less than 1 second nor more than 5 seconds. The duration of the test shall begin when 95 percent of the test potential is reached.

c. Points of application of test voltage: Between terminals.

d. Limiting value of surge current: Shall not exceed 50 milliamperes (mA) during charging and discharging.

e. Examination after test: Capacitors shall be examined for evidence of damage, arcing, and breakdown.

4.7.3 Barometric pressure (see 3.7). Capacitors shall be tested in accordance with method 105 of MIL-STD-202. The following details shall apply:

a. Method of mounting: Not applicable.

b. Test condition: D (100,000 ft).

c. Tests during subjection to reduced pressure: A test potential of 100 percent of rated dc voltage for style CMS04, 500 volts and 150 percent for all other styles and voltages shall be applied between the terminals for 60 +15, -0 seconds. Surge current shall not exceed 50 milliamperes.

d. Examination after test: Capacitors shall be examined for evidence of damage, arcing, and breakdown.

4.7.4 Insulation resistance.

4.7.4.1 At room ambient temperature (see 3.8.1). Capacitors shall be tested in accordance with method 302 of MIL-STD-202. The following details shall apply:

a. Test condition: A. (At the option of the manufacturer, a higher voltage may be used.)

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b. Points of measurement: From terminal to terminal. (Condensed moisture may be removed by a blast of air.)

c. Electrification time: Not less than 10 seconds and not more than 2 minutes.

4.7.4.2 At high ambient temperature (see 3.8.2). Capacitors shall be subjected to the high ambient temperature, 125 +5, -0°C for a periodic of time sufficient to reach thermal stability and shall be measured as specified in 4.7.4.1.

4.7.5 Capacitance (see 3.9). Capacitance shall be measured in accordance with method 305 of MIL-STD-202. The following details shall apply:

a. Test frequency: 1 megahertz (MHz) ±1,000 hertz (Hz) when the nominal capacitance is 1,000 pF or less, and 1 kilohertz (kHz) ±100 Hz when the nominal capacitance is greater than 1,000 pF. At the option of the manufacturer, capacitance measurements may be made at any frequency from 1 kHz to 1 MHz and referred to measurements at 1 MHz and 1 kHz, as applicable.

b. Limit of accuracy: Shall be ±0.2 percent of nominal capacitance value or ±0.2 pF, whichever is greater.

4.7.6 Dissipation factor (see 3.10). Dissipation factor shall be measured at a frequency of 1 MHz ±1,000 Hz when the nominal capacitance is 1,000 pF or less, and 1 kHz ±100 Hz when the nominal capacitance is greater than 1,000 pF. Measurement accuracy shall be within ±2 percent for dissipation factor and within ±5 Hz for frequency.

4.7.7 High voltage stabilization (see 3.11). The capacitors shall be subjected to 200 percent of rated dc voltage at a minimum temperature of 125°C for a minimum of 168 hours. During the test, the capacitors shall be adequately protected against voltage surges of 25 percent or more of test voltage and surge current shall not exceed 50 ma. Capacitors shall be examined for evidence of damage, arcing, and breakdown. If over 5 percent of the capacitors fail during this test (any part blowing a fuse or tripping the protective device shall be considered a failure), the lot shall be rejected.

4.7.8 Solderability (see 3.12). The capacitors shall be tested in accordance with method 208 of MIL-STD-202. The following details shall apply:

a. Number of terminations of each part to be tested: Two.

b. Depth of immersion in flux and solder: To a point at which exposed metal is solderable.

4.7.9 Vibration, high frequency (see 3.13). Capacitors shall be tested in accordance with method 204 of MIL-STD-202. The following details and exceptions shall apply:

a. Mounting: Capacitors shall be rigidly mounted by the body to a vibration-test apparatus.

b. Test condition: B.

c. Measurements during vibration: During the last cycle in each direction, an electrical measurement shall be made to determine intermittent contacts of 0.5 ms or greater duration or open- or short-circuiting.

d. Examination and measurements after test: Capacitors shall be examined for evidence of mechancial damage.

4.7.10 Temperature coefficient and capacitance drift (see 3.14).

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4.7.10.1 For qualification inspection. Capacitance measurements shall be made in accordance with 4.7.5 (a frequency of 100 kHz ±10 percent may be used as an alternate) at the following temperatures in the order listed:

Step	Temperature °C	Step	Temperature °C
1	25 ±2	6	45 ±2
2	-55 +0, -2	7	65 ±2
3	-40 ±2	8	85 ±2, -0
4	-10 ±2	9	125 +2, -0
5	25 ±2	10	25 ±2

The measurement at each temperature shall be recorded when two successive readings taken at 5-minute intervals at that temperature indicate no change in capacitance. An accuracy of ±0.25 percent of nominal capacitance +0.05 pF shall be maintained for measurement of capacitance change.

4.7.10.1.1 Temperature coefficient. The temperature coefficient shall be computed as follows:

$$TC = \frac{(C_X - C_5)\ 10^6}{(T_2 - T_1)\ C_5}$$

Where:

TC = Temperature coefficient in parts per million per degree centigrade.
C_X = Capacitance at test temperature in pF.
C_5 = Capacitance at step 5 temperature in pF.
T_1 = 25°C.
T_2 = Test temperature in degrees C.

4.7.10.1.2 Capacitance drift. Capacitance drift shall be computed by dividing the greatest single difference between any two of the three values recorded at 25°C by the step 5 value recorded at 25°C.

4.7.10.2 For quality conformance inspection. Capacitance measurements shall be made as specified in 4.7.10.1, except that measurements shall be made only for steps 1, 2, 5, 9 and 10.

4.7.10.3 Continuous-curve temperature coefficient. As an alternate to the measurements specified in 4.7.10.1, a continuous curve of capacitance versus temperature may be produced by subjecting the capacitors to a slowly varying temperature. The temperature shall be varied from 25°C to -55°C, to 125°C, and to 25°C. A temperature-sensing device shall be embedded in a dummy capacitor in a manner to assure accurate internal readings in the capacitor under test. The temperature shall be varied slowly enough to produce a smooth, uniform curve with no loops at -55°C or 125°C. Measurements shall be made at a frequency of 100 kHz ±10 kHz. Accuracy shall be as specified in 4.7.10.1.

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4.7.11 Thermal shock and immersion (see 3.15).

4.7.11.1 Thermal shock. The capacitors shall be tested in accordance with method 107 of MIL-STD-202. The following details and exceptions shall apply:

a. Test condition: A, for group A and A-1, for qualification and group C, except that step 3 shall be 125°C +3°C, -0°C.

b. Measurement before cycling: Capacitance.

c. Measurement after cycling: Not applicable.

4.7.11.2 Immersion. The temperature cycling shall be followed by immersion cycling in accordance with method 104, test condition B, of MIL-STD-202. The capacitors shall be visually examined for evidence of deterioration, permanent damage to leads and case, and corrosion on the leads. Dielectric withstanding voltage, insulation resistance, capacitance, and dissipation factor shall then be measured as specified in 4.7.2, 4.7.4.1, 4.7.5, and 4.7.6, respectively.

4.7.12 Shock, specified pulse (see 3.16). The capacitors shall be tested in accordance with method 213 of MIL-STD-202. The following details shall apply:

a. Mounting: The capacitors shall be rigidly mounted by the body.

b. Test condition: I (100 G (pk), sawtooth).

c. Measurements during shock: During the last shock in each direction, an electrical measurement shall be made to determine intermittent contacts of 0.5 ms or greater duration or open- or short-circuiting.

d. Measurements after shock: Capacitors shall be examined for evidence of mechanical damage.

4.7.13 Terminal strength (see 3.17).

4.7.13.1 Pull test. All terminals shall be subjected for 5 ±1 seconds, to pull of 5 pounds +2, -0 ounces in a direction perpendicular to the leads.

4.7.13.2 Twist test. The capacitors shall be tested in accordance with method 211, test condition D, of MIL-STD-202.

4.7.14 Moisture resistance (see 3.18). Capacitors shall be tested in accordance with method 106 of MIL-STD-202. The following details and exceptions shall apply:

a. Number of cycles: The capacitors shall be subjected to twenty continuous temperature cycles.

b. Mounting: The capacitors shall be mounted by the leads in a manner that will keep the capacitor body from touching the test fixture during test or measurement.

c. Conditioning prior to test: The capacitors shall be tested as specified in 4.7.11.1.

d. Polarization: A dc potential of 100 volts shall be applied across the capacitor terminals of 50 percent of the units under test and no voltage applied to the other 50 percent.

e. Vibration: The vibration requirement of steps 7a and 7b shall not apply.

f. Final measurements: After completion of step 6 of the final cycle, the capacitors shall be maintained at 25°C ±5°C, and 50 ±5 percent relative humidity for a period of 4 to 24 hours. Insulation resistance, dielectric withstanding voltage, capacitance, and dissipation factor shall then be measured as specified in 4.7.4.1, 4.7.2, 4.7.5, and 4.7.6, respectively. The capacitors shall then be examined for evidence of deterioration, permanent damage to the leads or case, and corrosion on the leads.

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4.7.15 Life. The capacitors shall be preconditioned for life test by subjecting them to a temperature of -55°C +0°C,-10°C for a minimum period of 48 hours. The capacitors shall then be subjected to the tests specified in 4.7.15.1 or 4.7.15.2, as applicable.

4.7.15.1 For qualification inspection (2,000 hour) (see 3.19). The capacitors shall be subjected to 150 percent of rated dc voltage at 125°C ±3°C for 2,000 +72, -0 hours. During the test, the surge current shall be limited to 50 milliamperes. At the conclusion of the test, capacitors shall be returned to the inspection conditions specified in 4.3.1, and shall be visually examined for evidence of mechanical damage; the dielectric withstanding voltage, insulation resistance, capacitance and dissipation factor shall then be measured as specified in 4.7.2, 4.7.4, 4.7.5 and 4.7.6, respectively (see 3.19).

4.7.15.2 For quality conformance inspection (see 3.19).

4.7.15.2.1 250 hours (group B inspection). The capacitors shall be tested as specified in 4.7.15.1, except that the duration of the test shall be 250 +48, -0 hours. The measurements specified in 4.7.15.1 shall be made at the conclusion of 250 hours of life test.

4.7.15.2.2 2,000 hours (group C inspection). All group B life samples shall remain on test for an additional 1750 +72, -0 hours. The measurements specified in 4.7.15.1 shall be made at the conclusion of 2,000 hours of life test.

4.7.16 Fungus (see 3.20). The capacitors shall be tested in accordance with method 508 of MIL-STD-810. Pretest and post test measurements are not required.

4.7.17 Resistance to solvents (see 3.21). The capacitors shall be tested in accordance with method 215 of MIL-STD-202. The following details shall apply:

a. Portion of specimen to be brushed: That portion on which marking is present.

b. Number of specimens to be tested: As specified in table IV.

c. Permissible extent of damage: As specified in 3.21.

4.7.18 Resistance to soldering heat (see 3.22). The capacitors shall be tested in accordance with method 210 of MIL-STD-202. The following details and exceptions shall apply:

a. Depth of immersion: Within .250-inch of the case.

b. Test condition: B (260°C ±5°C; 10 ±1 seconds).

c. Cooling time prior to final measurement: 1 minute ±10 seconds.

d. Measurements after test - Insulation resistance, capacitance and dissipation factor shall be measured as specified in 4.7.4, 4.7.5, and 4.7.6, respectively.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-C-39028. In addition to any special marking required by the contract or purchase order (see 6.2), each unit pack shall be marked with the source and date codes (see 3.23).

6. NOTES

6.1 Intended use. The capacitors covered by this specification are intended for use in space, launch vehicles, or other high reliability applications.

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6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Title, number, and date of the applicable specification sheet, and the complete part number (see 1.2.1 and 3.1).

c. Packaging requirements for shipments from the capacitor manufacturer direct to the equipment manufacturer.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable qualified products list whether or not such products have actually been so listed by that date. The attention of the contractors is called to this requirement, and manufacturers are urged to have the products that they propose to offer to the Federal Government, tested for qualification, in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the qualified products list is the Department of the Air Force, Space Division/ALM; however, information pertaining to qualification of products may be obtained from the Defense Electronics Supply Center (DESC-E), 1507 Wilmington Pike, Dayton, Ohio 45444.

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APPENDIX A

PROCEDURE FOR QUALIFICATION INSPECTION

A10 SCOPE

A10.1 Scope. This appendix details the procedure for submission of samples, with related data, for initial qualification inspection of capacitors covered by this specification. The procedure for extending qualification of the required sample to other capacitors covered by this specification is also outlined herein. There shall be no combining of styles, each shall be qualified separately. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

A20 APPLICABLE DOCUMENTS.

This section is not applicable to this appendix.

A30 SUBMISSION

A30.1 Qualification based upon similar established reliability (ER) style. A manufacturer shall be considered qualified for each CMS style for which he is presently qualified to the "R" failure rate level of a similar ER style (see table VIII) and the following conditions:

a. Passes the qualifying activity audit.

b. Successfully completes group C inspection (table VIII) for the lowest (lowest value need not be less than 47 pF) and highest capacitance values for each different style manufactured.

TABLE VIII. Similar styles.

CMS style	Similar style
CMS04	CMR04
CMS05	CMR05
CMS06	CMR06

A30.2 Qualification not based upon similar ER style. A sample of the size required in table IV of the highest and lowest (lowest value need not be less than 47pF) capacitance value in each style for which qualification is sought shall be submitted.

A30.3 Test data. When inspections are to be performed at a Government laboratory, prior to submission, all sample units shall be subjected to all of the inspections indicated as nondestructive in table IV. Each submission shall be accompained by the test data obtained from these inspections. The performance of the destructive tests by the contractor on a duplicate set of sample units is encouraged, although not required. All test data shall be submitted in duplicate.

A30.4 Certification of material. When submitting samples for qualification, the contractor shall submit certification, in duplicate, that the materials used in his components are in accordance with the applicable specification requirements and the baseline documentation.

A30.5 Description of items. The contractor shall submit a detailed description of the capacitors being submitted for inspection including material, type of electrodes and lead metal, dielectric thickness, margin, etc.

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A40 EXTENT OF QUALIFICATION

A40.1 Restrictions upon qualification. Capacitance-range qualification will be restricted to values equal to and less than the capacitance value submitted. Capacitance-tolerance qualification will be restricted to tolerance equal to and wider than the tolerance submitted. DC rated voltage qualification will be restricted to that submitted. Characteristic qualification will be restricted to characteristics equal to and wider than the characteristic submitted.

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APPENDIX B

PROCEDURES FOR THE PRODUCT AND QUALITY AUDIT

B10 SCOPE

B10.1 Scope. This appendix provides guidelines to the certification team for the implementation of CMS certification of manufacturing facilities. These guidelines will establish a uniform approach by the certification team in the assessment of the manufacturer's capability to consistently produce CMS capacitors by providing instructions for assuring process control, applicable test methods and records. A checklist is included for the use of the certification team (see figure 3). This appendix is a mandatory part of this specification. The information contained herein is intended for compliance.

B20 APPLICABLE DOCUMENTS.

This section is not applicable to this document.

B30 INSTRUCTIONS

B30.1 Baseline documentation. At the time of the initial survey, a baseline shall be established. It is not intended that the manufacturer divulge proprietary information. Each manufacturer shall provide a detailed process flow chart which shall include as a minimum the following:

a. All major process steps and inspection points by descriptive title, document number and revision.

b. The identification of all critical manufacturing and inspection steps as determined by the manufacturer in conjunction with the audit team at the time of baseline approval.

B30.2 Changes to baseline documentation.

a. Minor changes. Any subsequent departure from the approved flow chart (i.e., additions or deletions) or inspections deemed critical are allowed provided they do not violate the changes prohibited by 30.2b (major changes). These minor changes shall be described in general terms to the audit team during subsequent audits. This action shall result in an update of the baseline document.

b. Major changes. No changes (i.e., substitutions) shall be allowed in the following areas without prior review and approval of the qualifying activity. In these areas, partial or total requalification may be required as determined by the qualifying activity.

1. Capacitor Design (dielectric thickness, number plates, end margin i.e., as permitted by the baseline).

2. Silver to mica attachment.

3. Assembly.

4. Termination and connections.

5. Impregnation.

6. Encapsulation.

7. Curing process.

8. Solder formulation.

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APPENDIX B

B30.3 Audits. An initial audit and subsequent periodic audits (every 2 years minimum). The audit shall include, but not be limited to all of the items on the checklist plus MIL-STD-790 requirements. The result of any additional items that are assessed shall be added to the checklist for record purposes.

a. The audit team shall review the product assurance program and the manufacturing flow chart for adequacy and completeness prior to the audit of the manufacturing facilities and should be used as a baseline/reference during the course of the audit.

b. The audit team shall verify that all measuring and test equipment is functioning and calibrated. They shall also verify that test measuring procedures and equipment are sufficiently accurate for the required test.

c. The audit team shall check 12 months of records for the item being recertified. In addition, the location of the equipment and the adequacy of the operating procedure shall be checked and whenever practical, operations utilizing the equipment and procedures witnessed.

B30.4 General. The audit team shall insure that adequate documentation, trained personnel, equipment and visual aids are available to perform the inspections as shown on figure 3.

Check - Item	Inspection	Satisfactory
A. Flow Chart		
1. Major processes identified with revision		
2. Major inspection identified with revision		
3. Non-proprietary documents reviewed		
4. Screens/inspection points identified		
B. Baseline - shall describe as a minimum the following:		
1. Capacitor design by part number indicating the following with the necessary tolerance defined:		
dielectric thickness		
number of electrodes		
size		
2. Silver to mica attachment		
3. Assembly		
4. Termination and connections		
5. Impregnation		
6. Encapsulation		
7. Curing process		
8. Solder formulation		
C. Lot control plan		
D. Traceability plan		
E. Training plan for QC personnel		
F. Non-destructive test plan (group A)		
G. Destructive test plan (group B & C)		
H. Verification of MIL-STD-790 requirements		
I. Equipment calibration		
J. Record maintenance		
K. Reporting plan		

FIGURE 3. Certification check list.

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